ELECTRIC VEHICLE ANALYSIS

Revenue
392.03bn

CAGR
13.56%

Total EV sold
2.07M

Penetration Rate
3.61%

Total Vehicle sold
57.22M

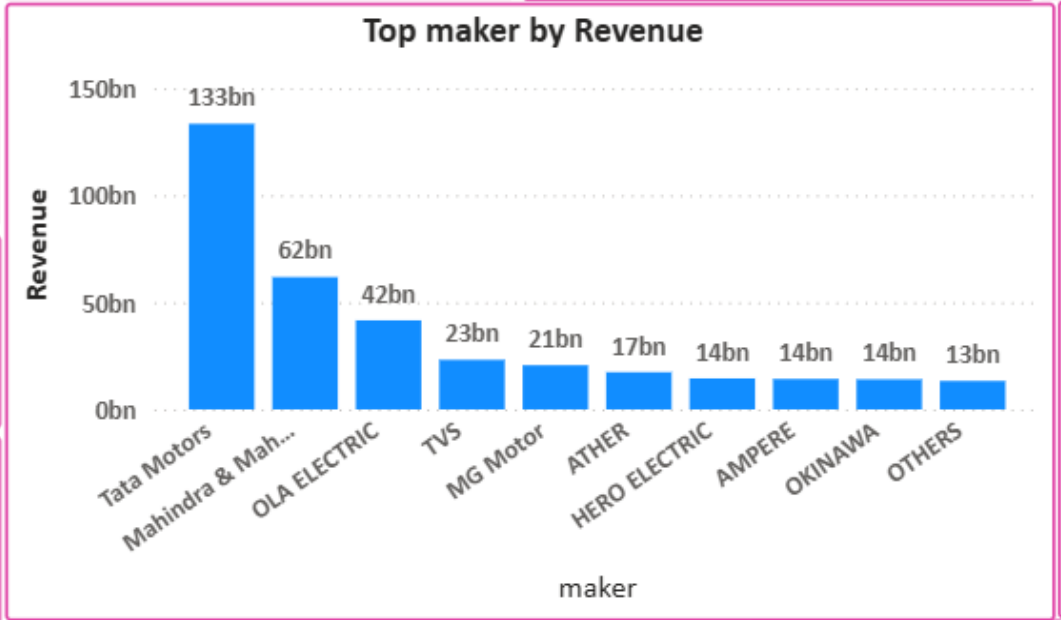
Maker wise CAGR
116.28%

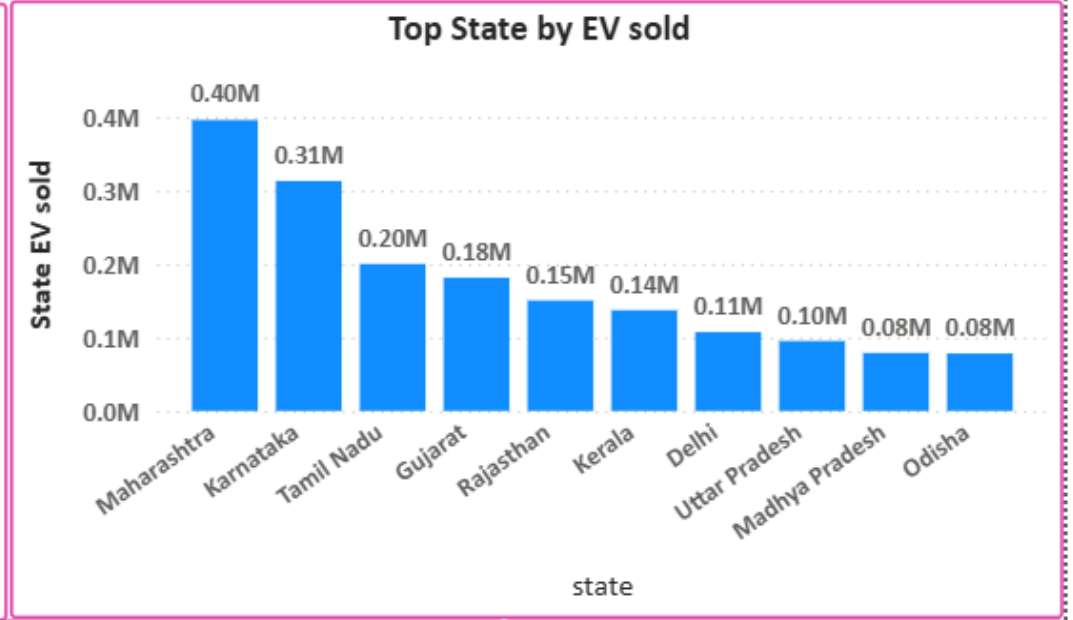
Vehicle category: All

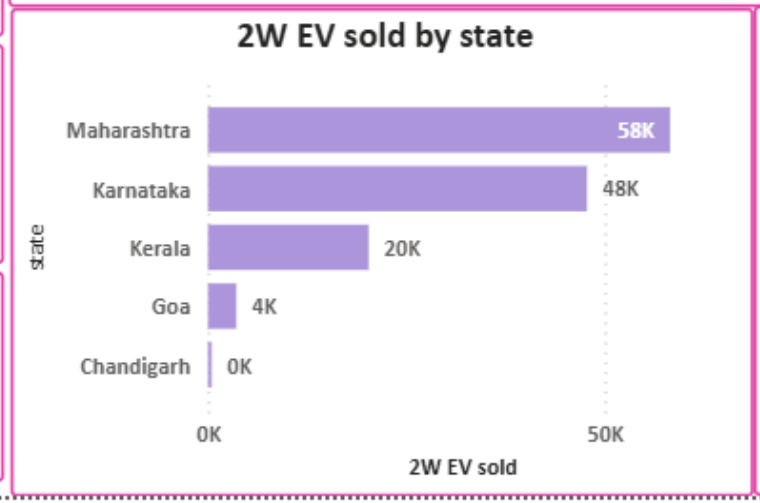
State: All

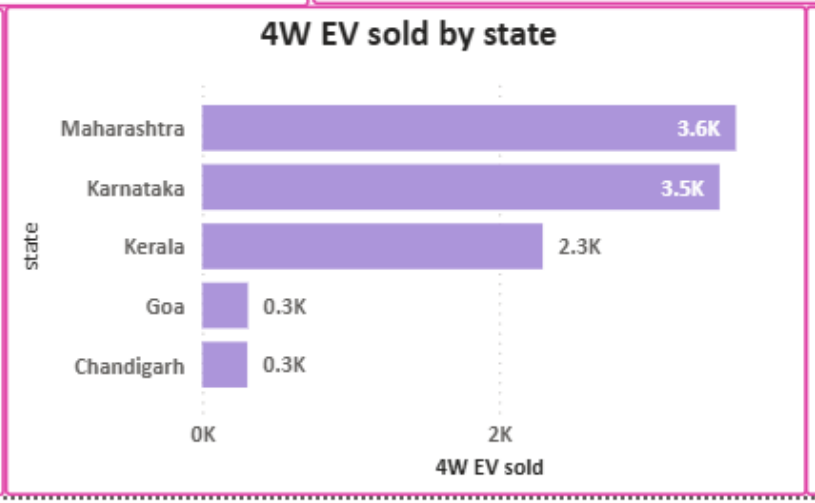
Maker: All

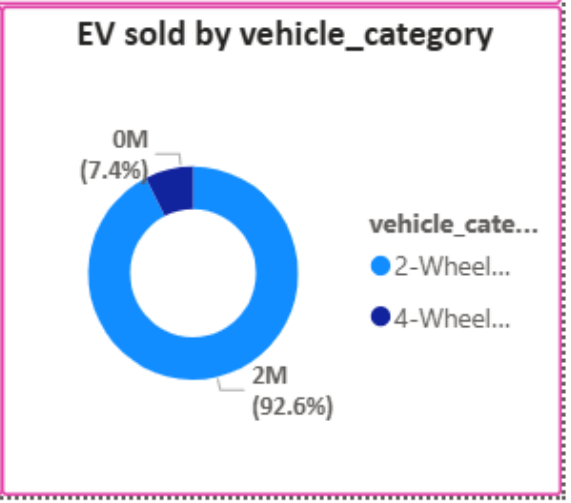
Year: All



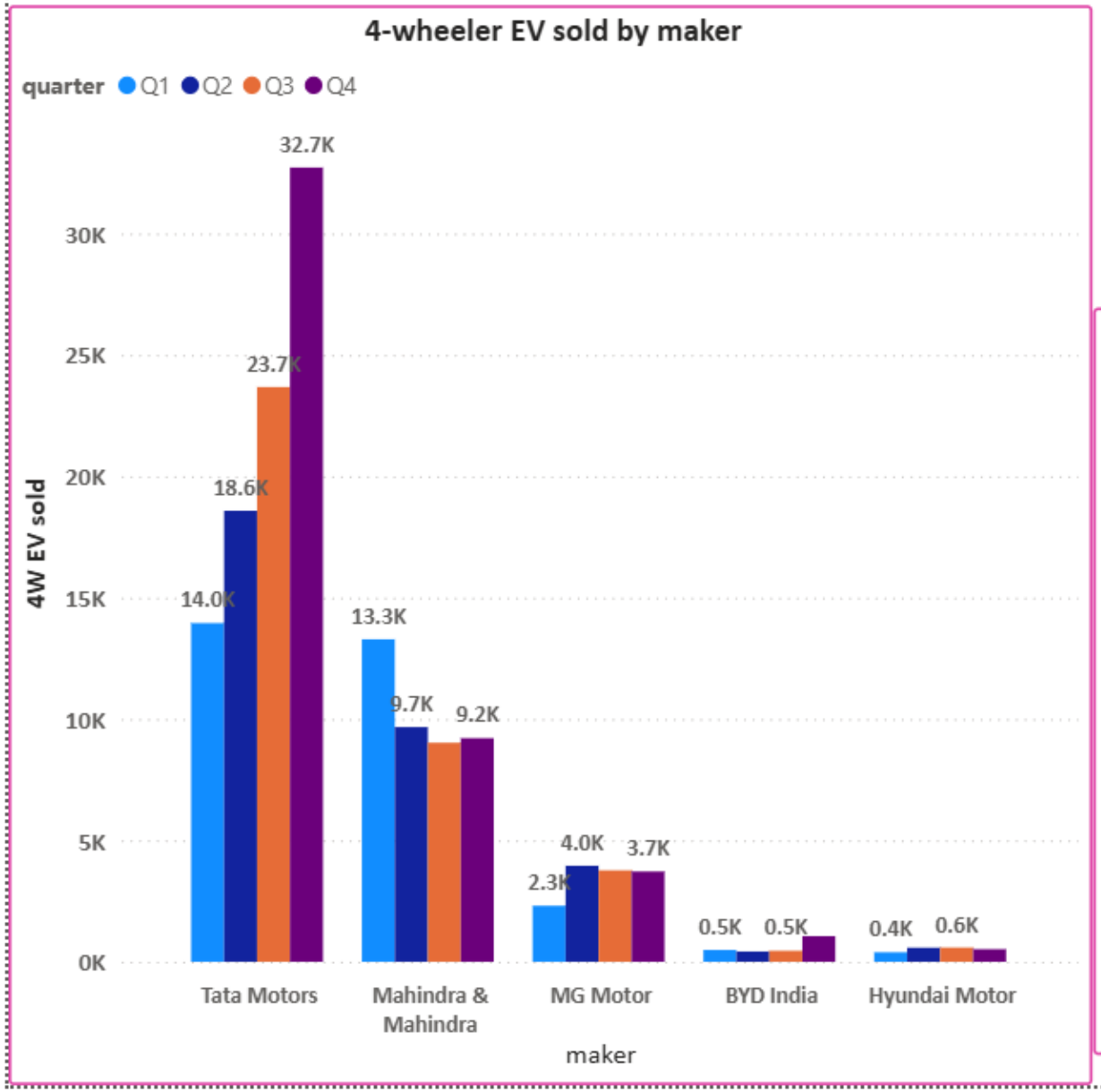




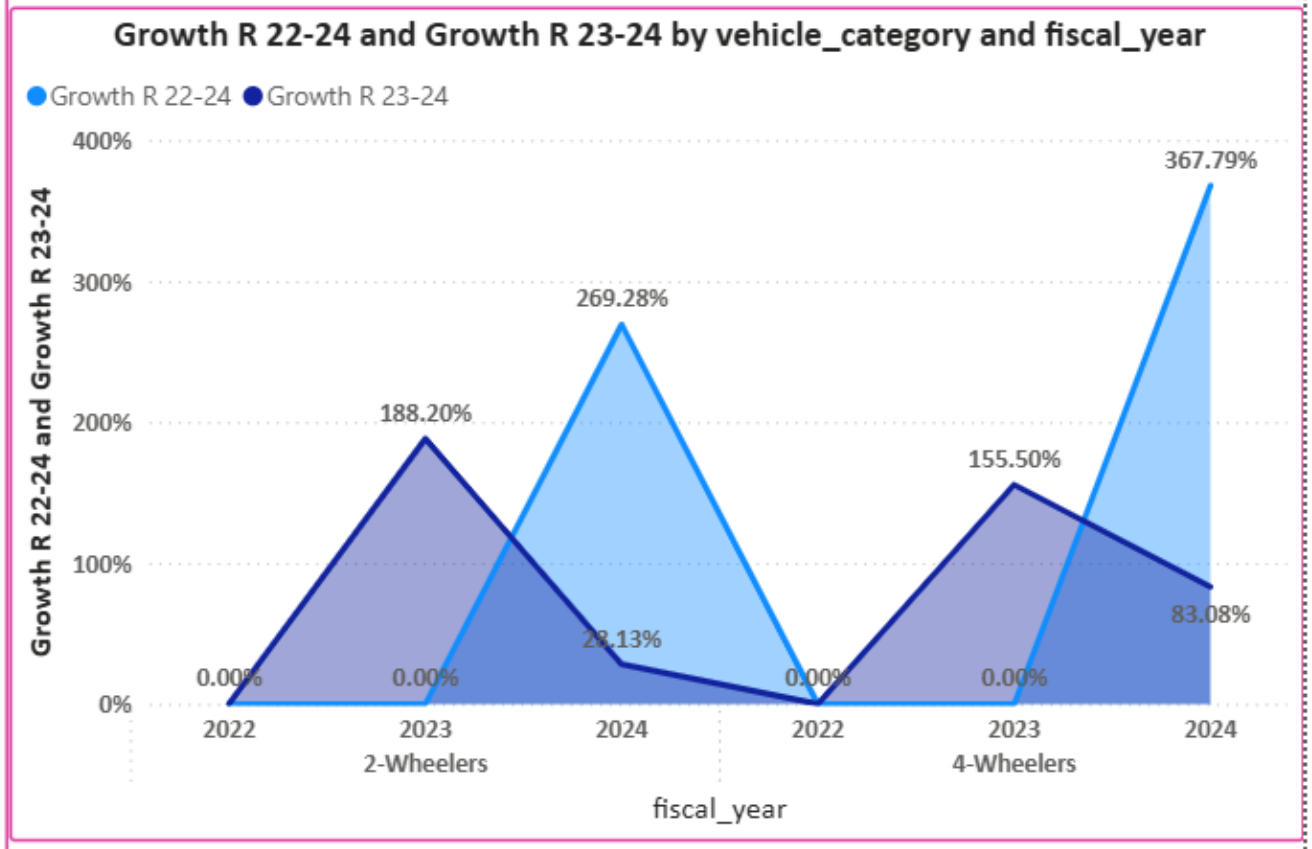




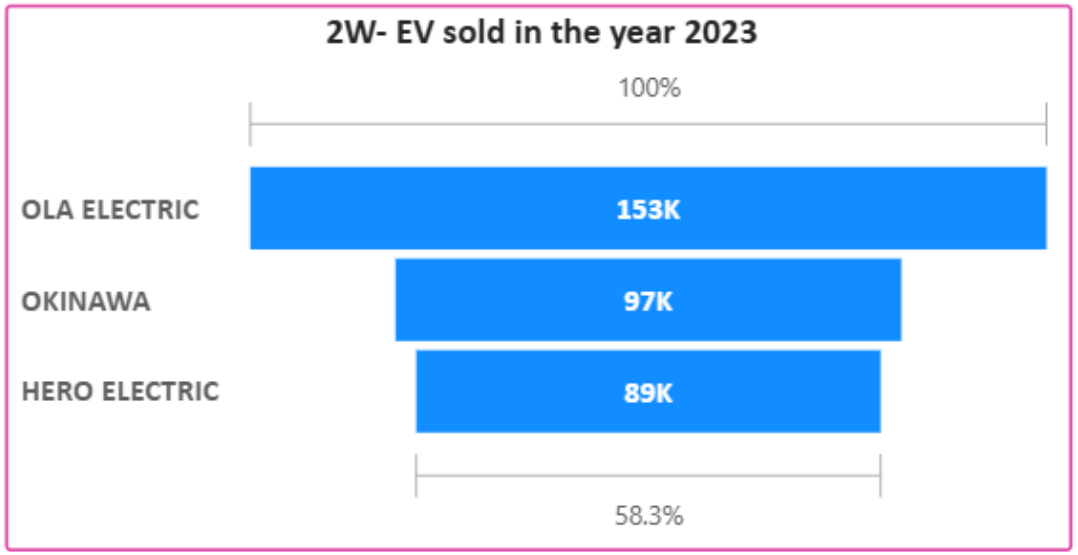
Quarter wise Electric 4-wheeler sold by Top 5 Makers

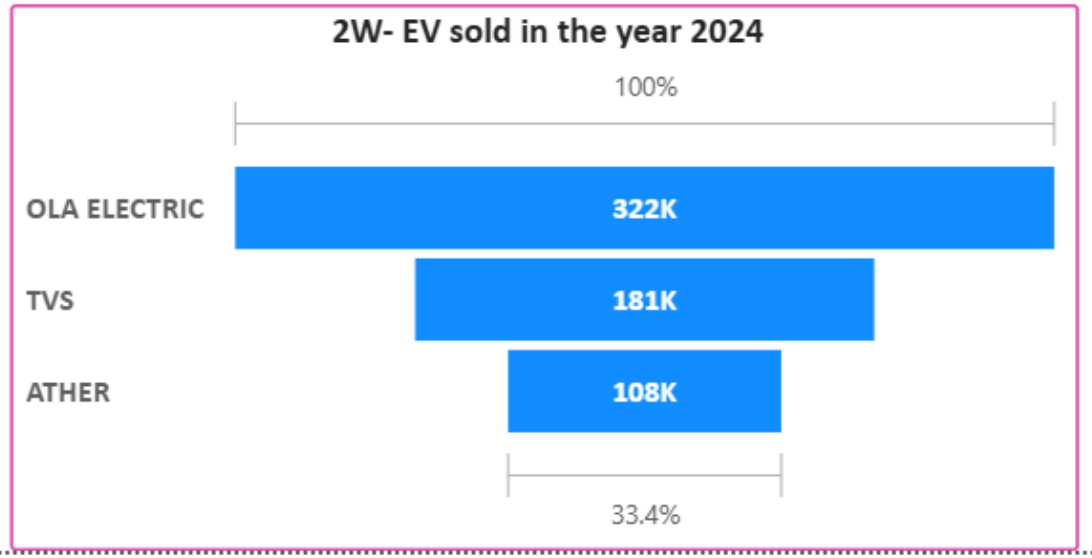


2- wheeler & 4-wheeler's Growth Rate by Year

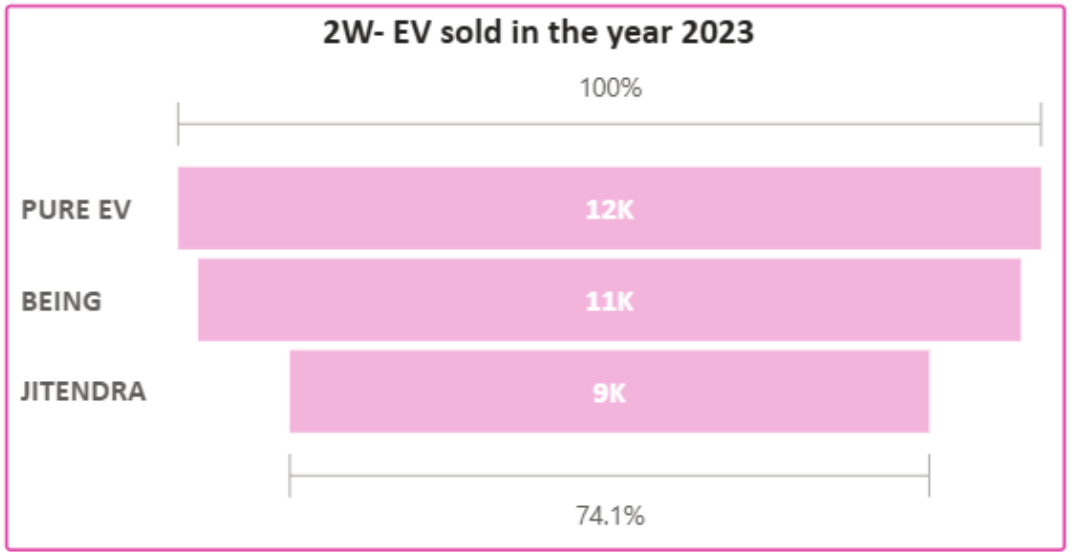


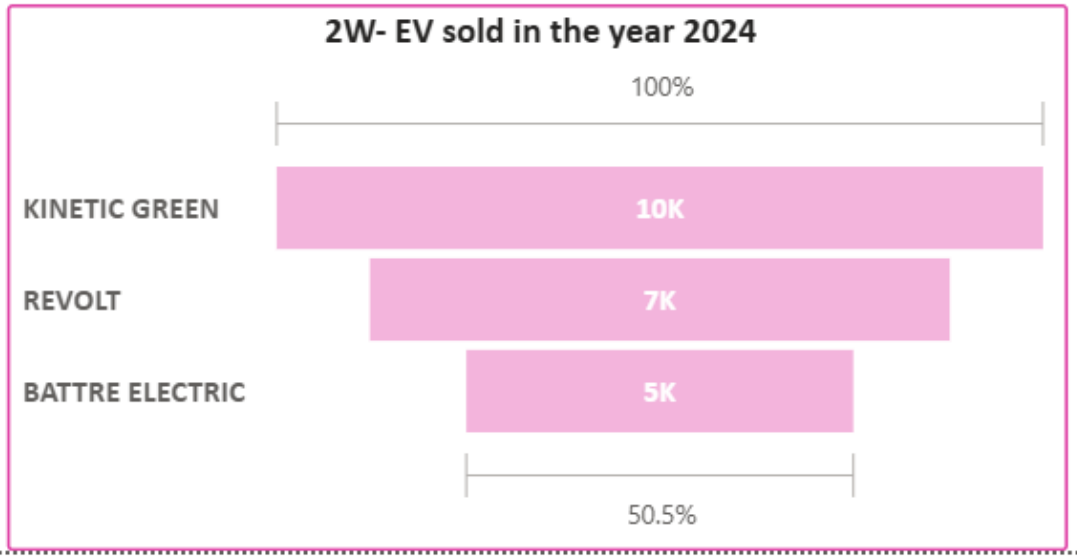
Top 3 Maker who sold Max electric 2-wheeler in year 2023 & 2024



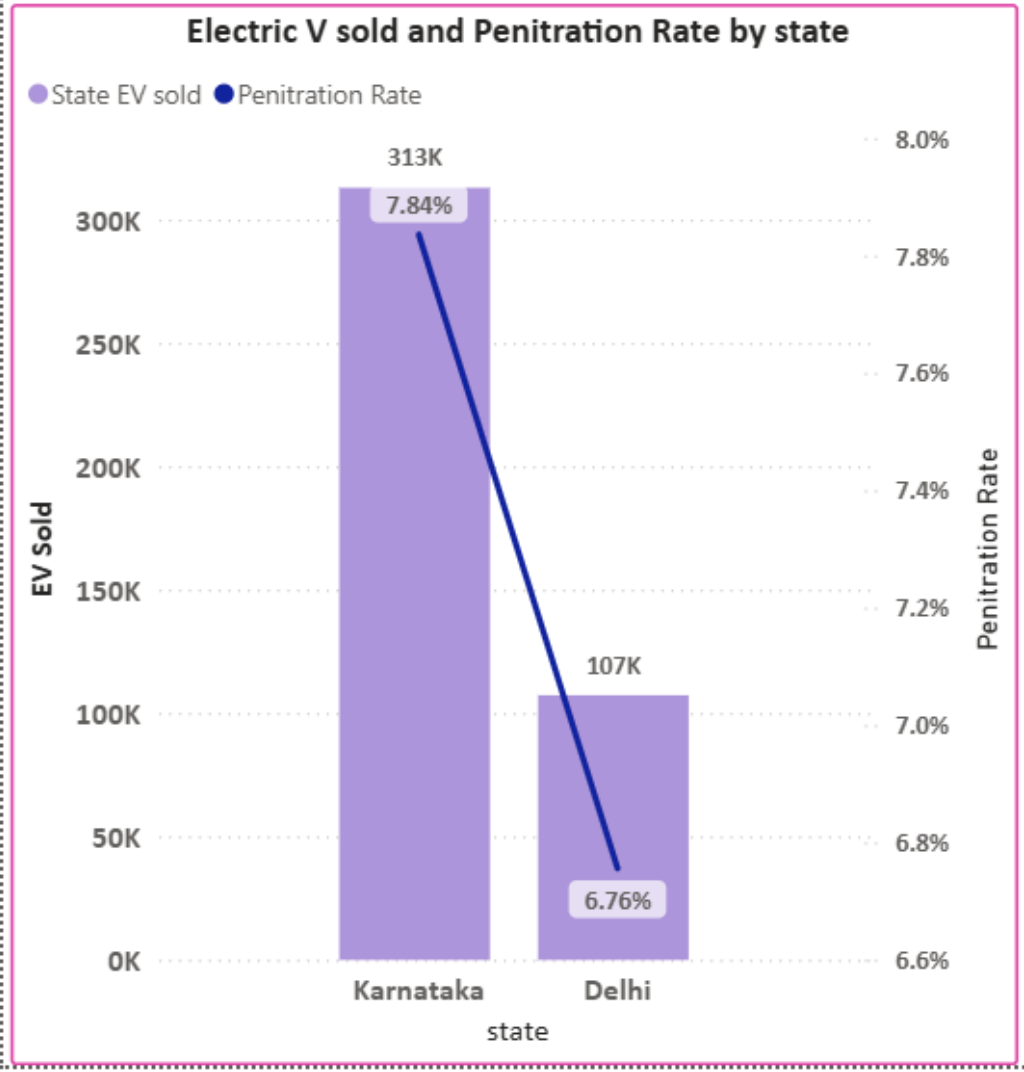


Bottom 3 Maker who sold electric 2-wheeler in year 2023 & 2024

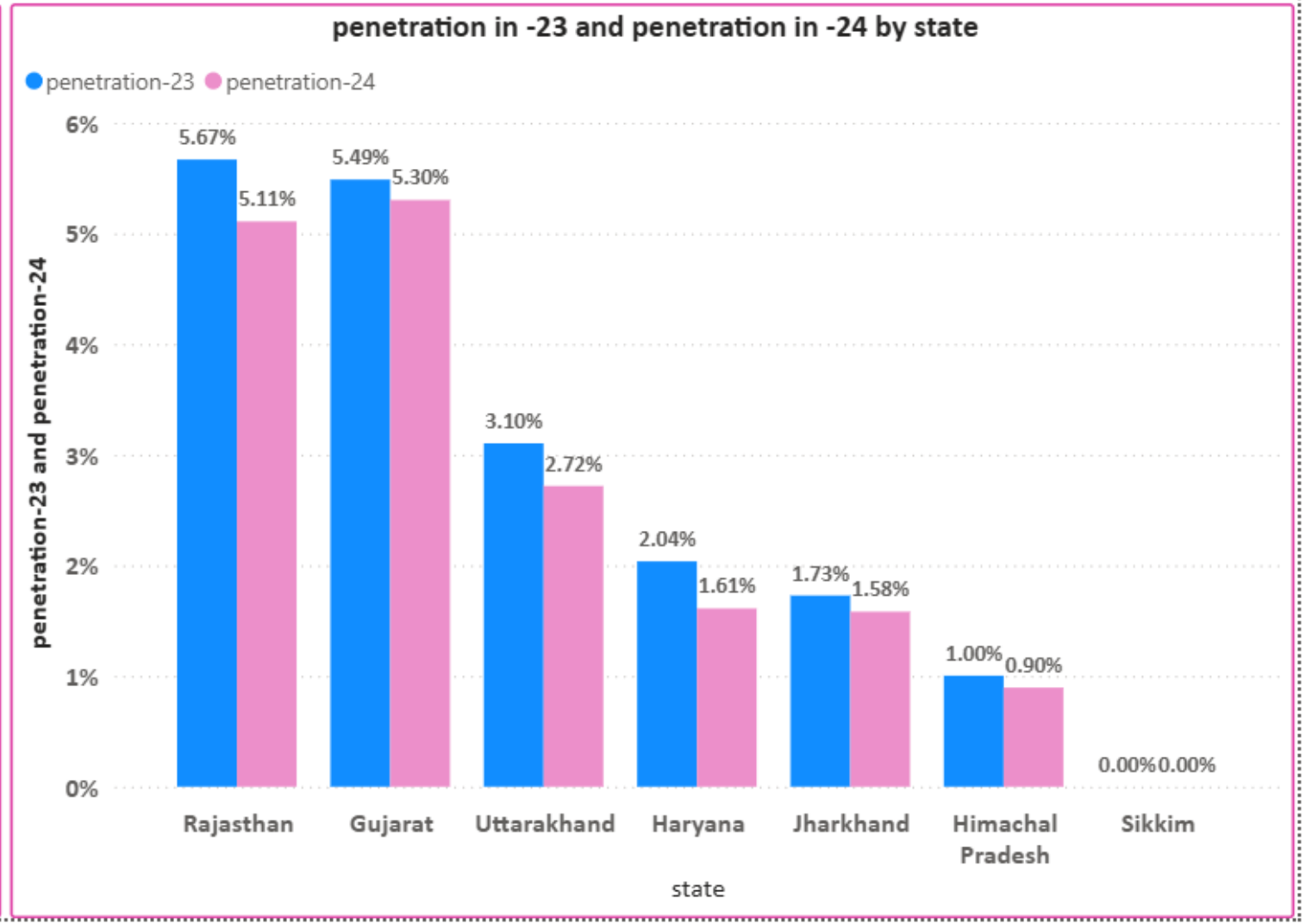


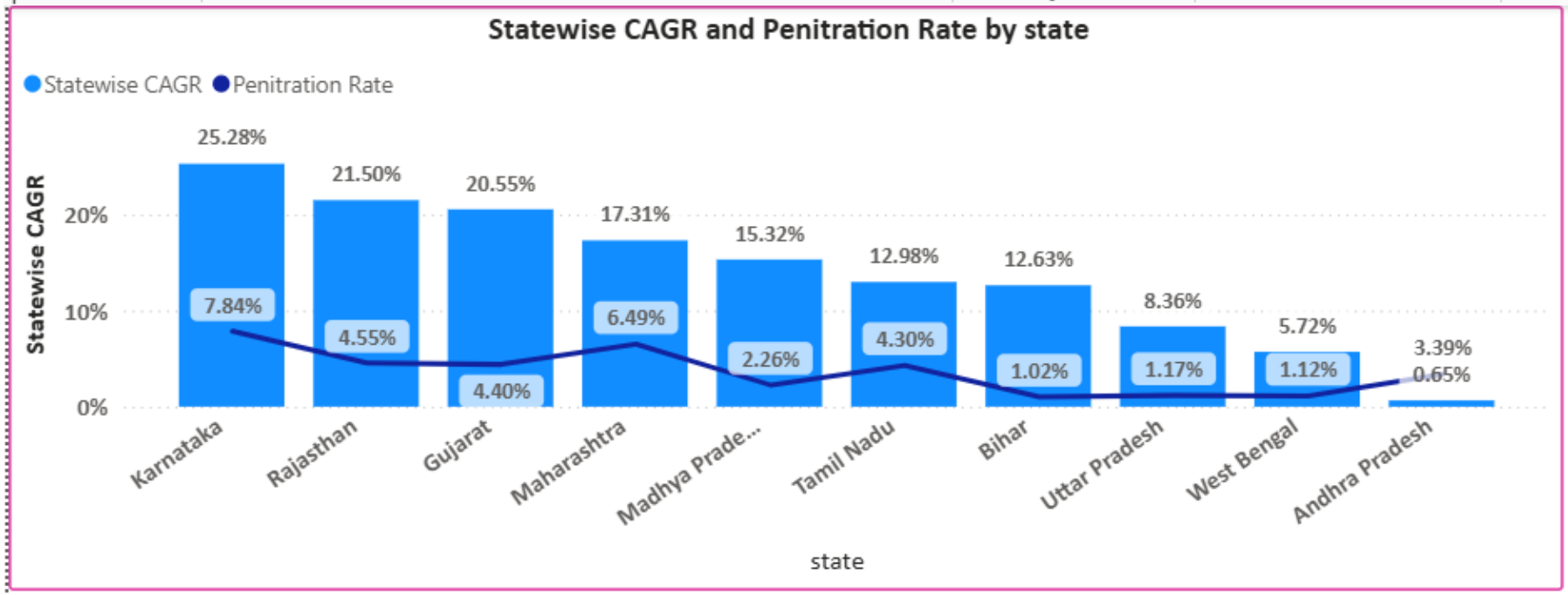


Comparison bet Delhi & Karnataka in terms of EV sold & Penetration Rate

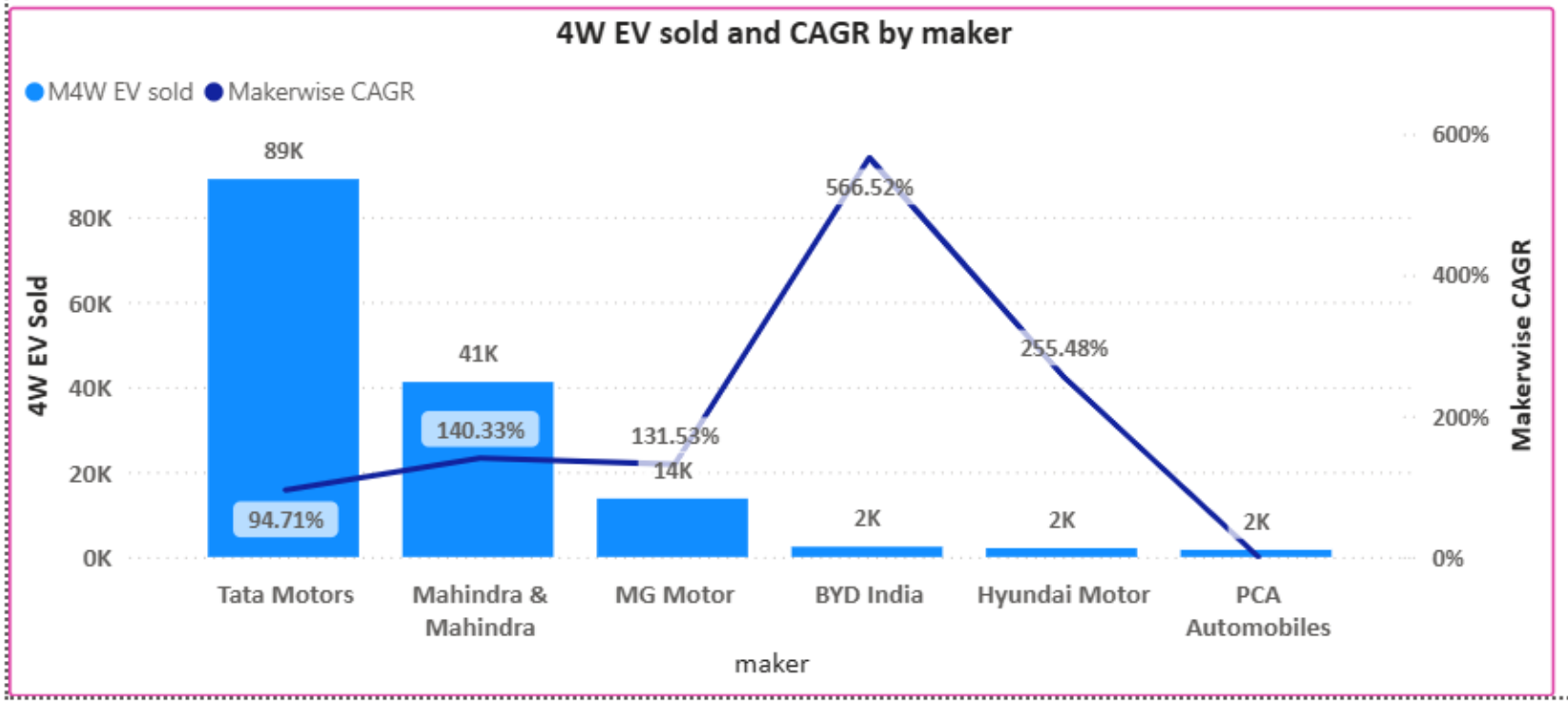


State which is having Penetration Rate decreases in the year 2024 from that of previous year -23

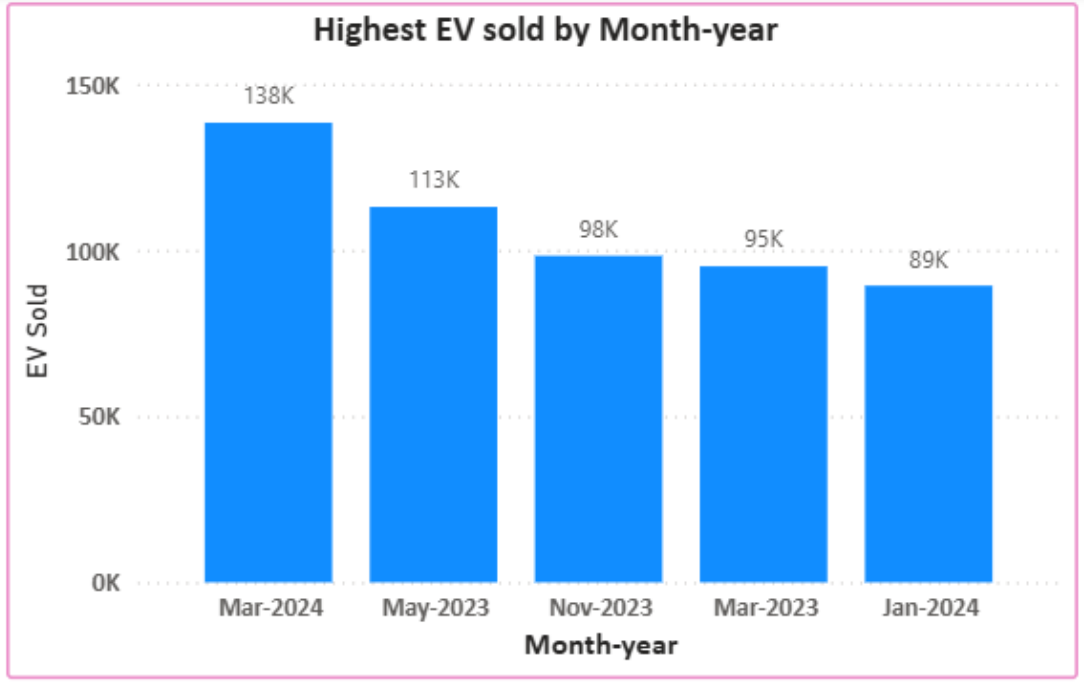


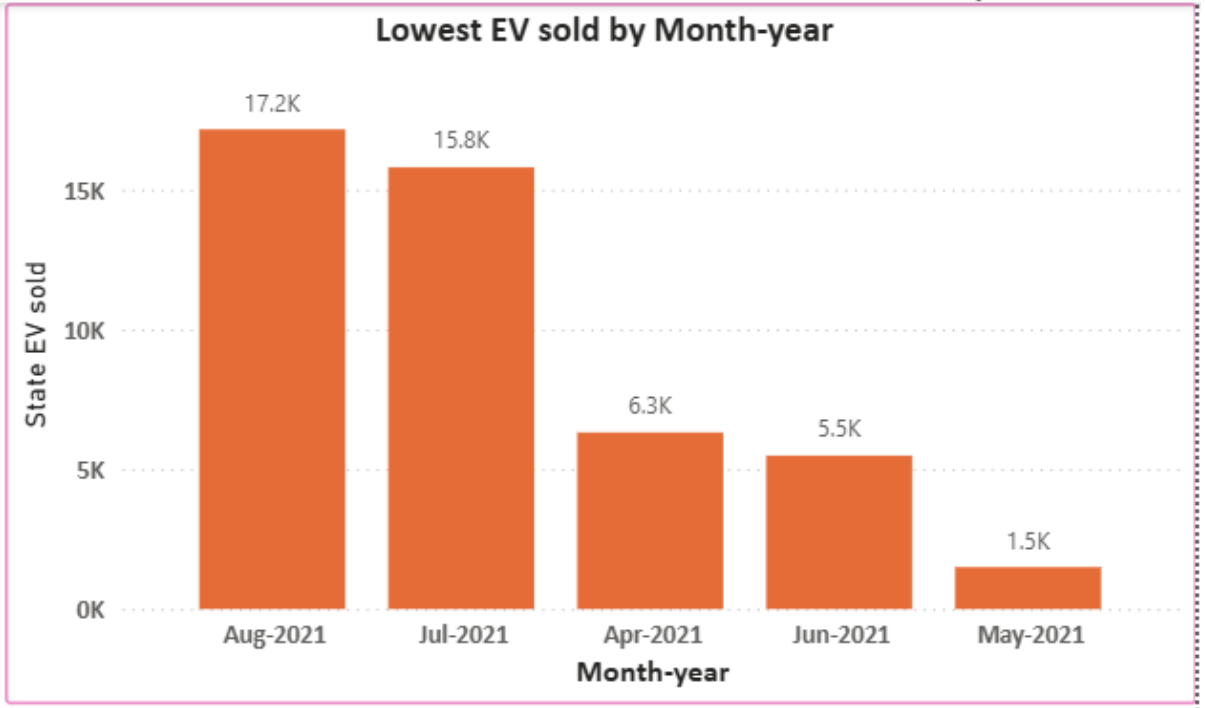


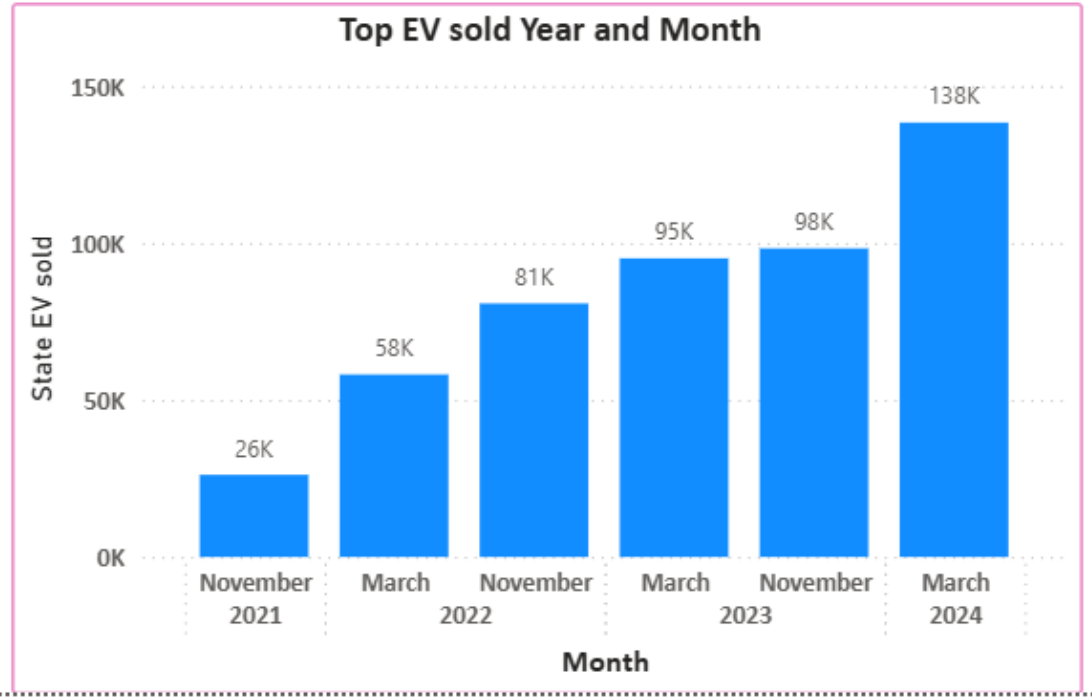
state	Statewise CAGR
Andhra Pradesh	0.65%
Bihar	12.63%
Gujarat	20.55%
Karnataka	25.28%
Madhya Pradesh	15.32%
Maharashtra	17.31%
Rajasthan	21.50%
Tamil Nadu	12.98%
Uttar Pradesh	8.36%
West Bengal	5.72%
Total	14.01%

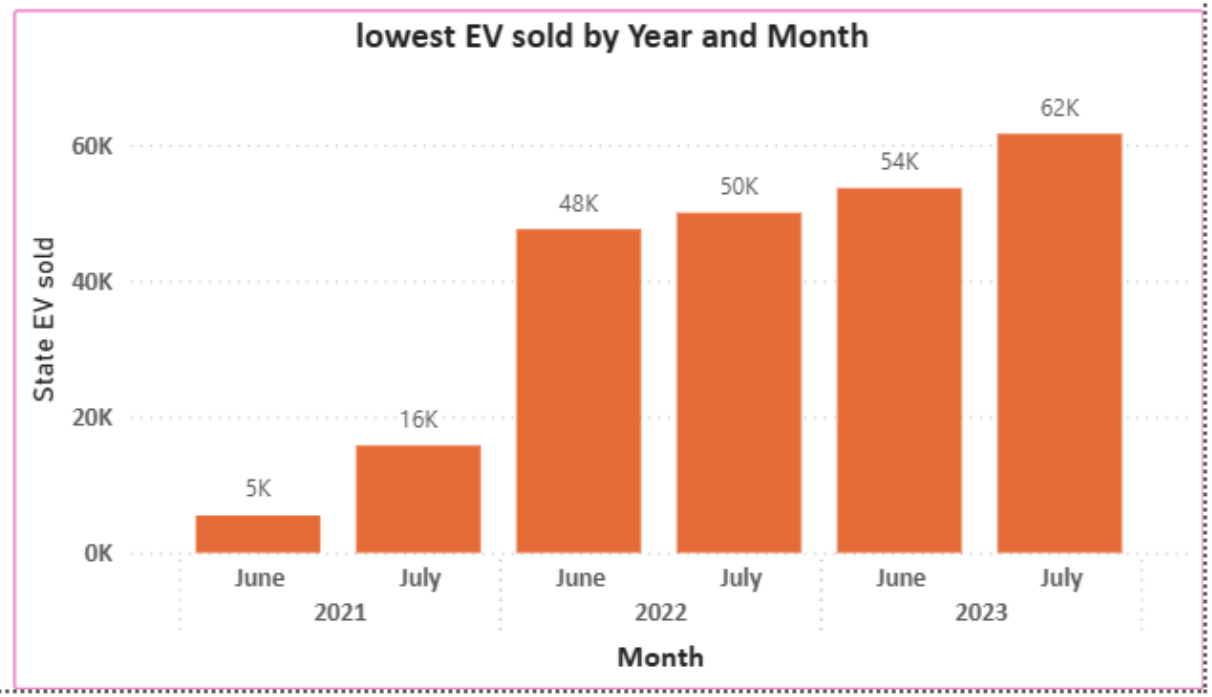


maker	Makerwise CAGR	M4W EV sold
BYD India	566.52%	2419
Hyundai Motor	255.48%	2076
Mahindra & Mahin...	140.33%	41193
MG Motor	131.53%	13753
Tata Motors	94.71%	88935


Highest EV sold by Month-year
150K
100K
50K
0K
EV Sold
138K
113K
98K
95K
89K
Mar-2024
May-2023
Nov-2023
Mar-2023
Jan-2024
Month-year


Lowest EV sold by Month-year
15K
10K
5K
0K
State EV sold
17.2K
15.8K
6.3K
5.5K
1.5K
Aug-2021
Jul-2021
Apr-2021
Jun-2021
May-2021
Month-year


Top EV sold Year and Month
150K
100K
50K
0K
State EV sold
26K
58K
81K
95K
98K
138K
November 2021
March
November
2022
March
November
2023
March 2024
Month

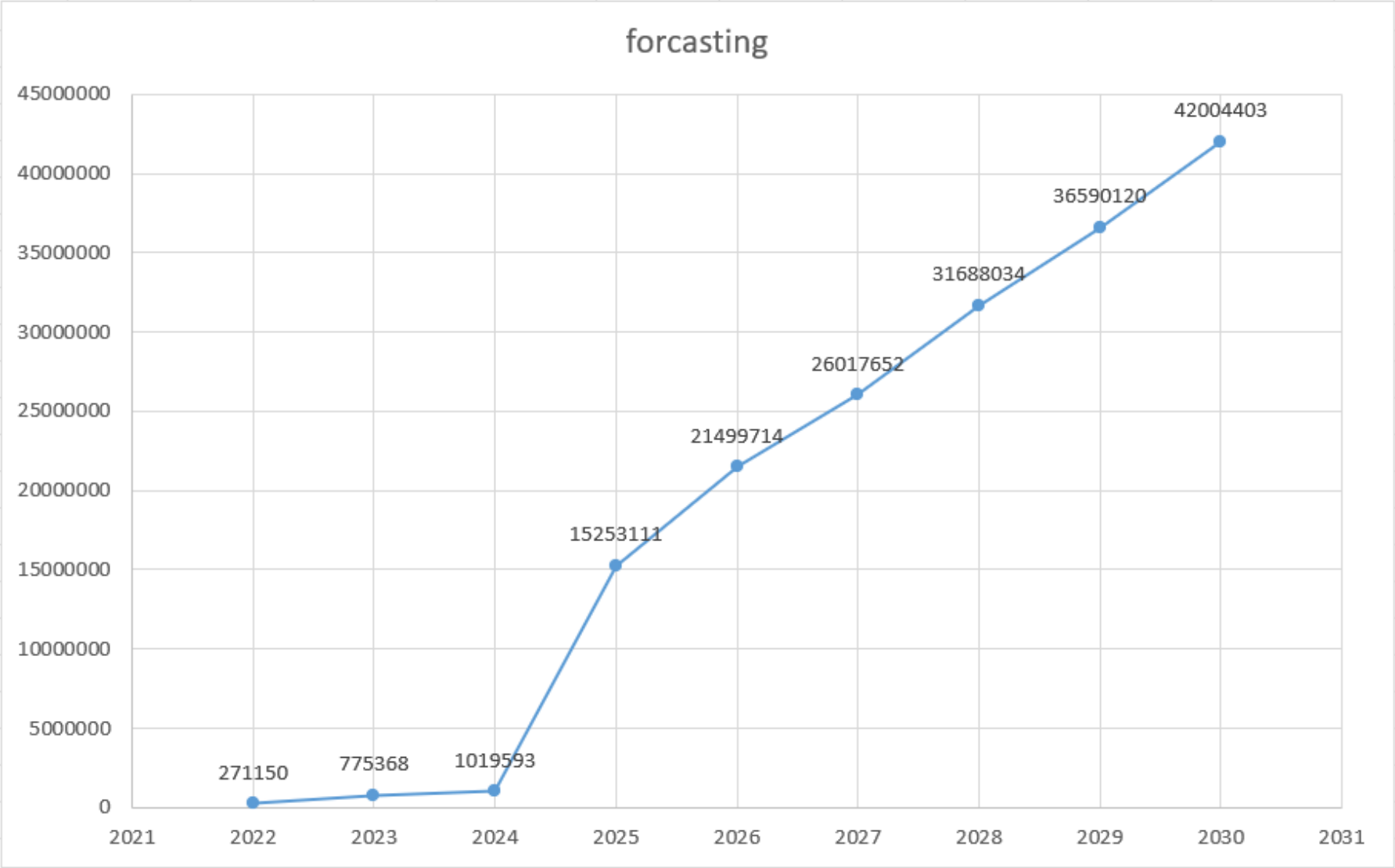

lowest EV sold by Year and Month
60K
40K
20K
0K
State EV sold
5K
16K
48K
50K
54K
62K
June
July
2021
June
July
2022
June
July
2023
Month

forcasting

year	EV sold	forcasting
2022	271150	
2023	775368	
2024	1019593	
2025	1437147	15253111
2026	1739148	21499714
2027	2118184	26017652
2028	2445864	31688034
2029	2807781	36590120
2030	3146873	42004403
	linear	based on CAGR

State with highest Penetration Rate

state	Statewise CAGR	State EV sold
Goa	27.41%	19684
Karnataka	25.28%	312995
Delhi	22.88%	107312
Rajasthan	21.50%	150366
Gujarat	20.55%	181389
Maharashtra	17.31%	396045
Odisha	13.54%	78267
Tamil Nadu	12.98%	200062
Chandigarh	10.53%	5279
Kerala	-3.80%	137060



Primary Reasons For Customers Choosing 4-wheeler EVs In 2023 And 2024:

- **Environmental concerns**: Many consumers are motivated by the desire to reduce their carbon footprint. EVS produce fewer emissions over their lifetime compared to ICE vehicles.

- **Government incentives**: Various government incentives make purchasing EVs more attractive. These can include tax credits, rebates, reduced registration fees, and access to carpool lanes. Such incentives help offset the higher initial purchase price of EVs, making them more competitive with traditional vehicles

- **Cost savings**: EVs offer significant savings in both fuel and maintenance costs. Electricity is generally cheaper than gasoline, and EVS have fewer moving parts, reducing the need for regular maintenance such as oil changes and spark plug replacements.

Government Incentives And Subsidies Impact On Adoption Rates Of 2-wheelers And 4-wheelers?

- **FAME II scheme**: The faster adoption and manufacturing of hybrid and electric vehicles (FAME II) scheme in India has played a pivotal role in promoting EV adoption. Under this scheme, electric two-wheelers received substantial support, achieving 75% of their revised target.
- **FAME II and production linked incentive (PLI) schemes**: While four-wheelers have also benefited from the FAME II scheme, the adoption rate has been slower, achieving only 55% of the revised target. High upfront costs and limited charging infrastructure remain significant barriers

State provides subsidies :-

Maharashtra provided the most subsidies for electric vehicles (EVS) in India. The state offer significant incentives to promote the adoption of EVs, including up to
1.5 lakh for four-wheeler ,
20,000 for two-wheelers, and
30,000 for three-wheelers

Gujrat , Karnataka, Meghalaya, Kerala and many more state provides good subsides in India.

Availability of charging stations infrastructure correlate with the EV sales

- **EV sales and penetration rates**: States with more developed charging infrastructure tend to see higher EV sales and penetration rates. For instance, Maharashtra, which has the highest number of charging stations, also shows significant EV adoption.
- **Challenges and solutions**: despite the progress, several challenges remain, such as the need for standardized connectors and reliable power supply. Addressing these issues is crucial for further accelerating EV adoption.

Recommendations:

India is ideal for EV sale business and to start its manufacturing unit

- ✓ **Charging Stations**: Collaborate with local governments, private players, and utility companies to develop a widespread and reliable charging infrastructure.
- ✓ **Target tier 1 & 2 cities**: it will led to capture high selling hub as every customers travel to the nearby basis.
- ✓ **Unique features & specifications**: incorporate unique features and technological innovations to differentiate your EVS from competitors. This could include advanced infotainment systems, autonomous driving capabilities, or enhanced safety features.
- ✓ Establish local manufacturing units to benefit from lower production costs, reduced tariffs, and government incentives.

- ✓ **Subsidies**: Maharashtra and many other state offers substantial incentives under the industrial promotion subsidy (IPS), particularly in less industrialized regions , where subsidies can reach up to 80% of the fixed capital investment
- ✓ **Ease of doing business**: India has a well- established industrial ecosystem, beneficial to automobile manufacturing
- ✓ **Stability and infrastructure:** Indian states has good infrastructure and a well-designed logistics network.
- ✓ **Penetration Rate**: As the penetration rate is 3.61 only. There are more scope for Electrical vehicle sales

Thank You